

ZILING ZHANG

Gen AI Architect / Data Scientist / Roboticist

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EXPERIENCE

Advisory Data Scientist

Automation Technical Implementation Group | Ricoh Graphic Communications

Aug 2022 – present Boulder, CO

Agentic AI

- Lead AI Coding Assist pilot with Claude Code for 280+ builders. Architect AWS Bedrock cloud infrastructure with SSO auth, per-user cost tracking, OpenTelemetry observability.
- Architect enterprise Gen AI transformation. Built Grace AI platform serving 280+ builders with Claude on AWS Bedrock, Docker microservices (Open WebUI, LiteLLM, vLLM), MCP tool integrations (Jira, Confluence, GitLab).
- Manage NVIDIA GH200×4 GPU cluster serving local LLMs, embedding, reranking, and image generation models, powering enterprise AI agents across multiple departments.
- Design Agentic Coding training programs. Hackathon judge, CU Boulder.
- Develop serverless ETL pipeline aggregating printer/sensor timeseries data. AWS Athena, Lambda. Supporting predictive maintenance inference with SageMaker serverless.

Physical AI

- Prototype Universal Robot (UR20) book palletizer. ROS2, computer vision, Cognex barcode scanner, FastAPI backend, React/Three.js digital twin (Cobot UI).
- Manage NVIDIA RTX Pro 6000 Blackwell cluster for simulation-based robotics and computer vision software CI pipelines.
- Hands-on expertise across NVIDIA's three-computer paradigm: GH200 (datacenter AI), RTX Pro 6000 Blackwell (Omniverse simulation), Jetson (edge robotics).
- Colorado School of Mines field session: Nvidia Cosmos robotic simulation data generation, Omniverse ray traced printshop digital twin.

Staff/Advisory Imaging Science Engineer

Boulder Development Section | Ricoh Graphic Communications

Aug 2019 – Aug 2022 Boulder, CO

Computer Vision + Automation R&D for Continuous Feed Industrial Printer

- Detect nozzle defect by training a semantic segmentation Deeplab v3+ neural network, accelerate with Nvidia TensorRT int8 quantization, achieve 37x speedup.
- Prototype nozzle uniformity algorithms, accelerate with SIMD and threading, achieve 40x speedup. Intel MKL/TBB, OpenCV, Visual Profiler, Concurrency Visualizer.
- Architect image processing pipeline that can withstand 7GB/s scanner data, perform realtime defect detection and compensation to achieve superior print quality.
- Standardize the team's coding practice, enforce modern C++ RAII style, C++ Core Guidelines, Google python style / pylint, modularize code into testable liblets.
- Setup CMake, google test, GitLab CI, docker runners for high velocity iteration.
- Filed 28 patents, granted 10, achieved Master Inventor title, fastest in Ricoh Boulder.

Robotics Research Assistant

Human Centered Robotics Lab | Colorado School of Mines

Aug 2018 – Dec 2019 Golden, CO

- DARPA subteranean challenge Mines team, SLAM LiDAR 3D mapping ROS node.
- Toyota InfoTechnology Center, V2X IoT multi-vehicle collaborative perception.
- Metcalf Archaeology, long term place recognition AR iOS app (Swift, OpenCV C++).
- Kinect + FLIR lpton sensor fusion, forehead temperature measurement TurtleBot.

SKILLS

Gen AI Stack

Claude Code Claude Agent SDK
Bedrock vLLM LiteLLM

Accelerated Compute

CUDA TensorRT Warp
CuTile MKL TBB SIMD

ROS Stack

Omniverse OpenCV Isaac Sim
Newton Jetson Python
C++20

Cloud Stack

AWS Serverless SageMaker

Frontend Stack

React TypeScript Three.js

EDUCATION

M.S. Computer Science

Colorado School of Mines

- Aug 2018 – Dec 2019
- Robotics, HPC ★ GPA 3.96
- Courses: Computer Vision, Parallel Computing, Robotics, Computer Architecture
- Paper: Collaborative Localization for Occluded Objects in Connected Vehicular Platform
4th author, 2019 IEEE 90th Vehicular Technology Conference

B.S. Applied Physics

University of Science and Technology of China

- Sep 2007 – June 2011
- Condensed Matter Physics
- Paper: Speed of Last Vehicle Feedback Strategy in Intelligent Transportation Systems
3rd author, International Journal of Modern Physics C, 2011

Scientific Translator (English -> Chinese)

Freelance Contracts

- 📅

Jan 2012 – Aug 2018

📍 Remote
- 💬

(NetEase Inc.) Translate OpenCourseWare, Coursera lectures subtitles, 1.87 million words.
- 📖

(Yeeyan.org) Book translation, Net Smart by Howard Rheingold, MIT press.

Software Developer

JindianTuteng Talent information solution Inc.

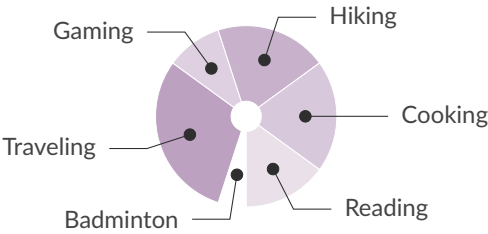
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Oct 2011 – June 2012

📍 Guangzhou, China
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Rule based transaction fraud detection for Foreign Exchange Management.
Java, J2EE, JSP, IBM WebSphere, IBM DB2.

INTERESTS



CONFERENCES

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SIGGRAPH 2024
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NVIDIA GTC 2025
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NVIDIA GTC 2026
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SIGGRAPH 2026